

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date	19 July 2026
Team ID	[Enter Your Team ID]
Project Name	BookEase – Multi-Role Online Bookstore Platform
Maximum Marks	4 Marks

Functional Requirements

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration through form (name, email, mobile, password) Email already-registered validation Password confirmation check
FR-2	User Login & Session	Login via email/password JWT-based session token Forgot password / reset password via token
FR-3	Profile Management	Update name, mobile, address Upload/update profile picture View order history from profile
FR-4	Book Browsing & Search	Browse by genre/category Search by title or author Filter by price, rating, language Sort by price, rating, latest Best-seller / new-arrival tags
FR-5	Cart Management	Add / update / remove items Apply coupon codes Cart persists across logout-login (cached + synced with server)
FR-6	Wishlist	Add / remove books from wishlist View saved books
FR-7	Checkout & Orders	Place order with shipping address Choose payment method (COD / online) View order status & invoice Cancel order with a mandatory reason
FR-8	Seller Operations	Seller registration with business details Add / update / delete own book listings View & fulfil orders containing own books
FR-9	Admin Operations	Approve / reject seller accounts Manage users (activate/suspend/delete) & set ratings Manage books & categories (any seller) Manage all orders & update status View dashboard analytics (revenue, orders by status)

Non-functional Requirements

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	Simple, responsive interface with clear, role-specific navigation for customers, sellers, and admins across desktop and mobile.
NFR-2	Security	JWT-based authentication, bcrypt password hashing, role-based access-control middleware, and server-side validation on every mutating request.
NFR-3	Reliability	Stock levels are validated and updated transactionally during checkout and order cancellation to prevent overselling or data inconsistency.
NFR-4	Performance	Paginated book listings, MongoDB text indexes on title/author/description, and client-side caching of the cart for instant rendering.
NFR-5	Availability	Stateless REST API that can run as multiple instances behind a load balancer; MongoDB Atlas provides managed replication.
NFR-6	Scalability	Layered MERN architecture allows the React frontend (Vercel/Netlify) and Express backend (Render/Railway) to scale independently of the MongoDB Atlas data tier.